

MANGALIA, Romania

WMO: 154990

Lat: 43.8161N

Lon: 28.5875E

Elev: 24

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	21.7	11.0	9.6	18.2	15.2	11.9	22.5	36.1	38.0	31.4	40.3	10.3	290	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	83.5	74.1	81.9	73.9	80.6	73.3	78.0	80.8	76.6	79.9	75.3	78.8	7.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.8	80.0	75.6	134.0	78.9	74.0	127.1	77.8	41.6	80.6	40.1	80.2	38.8	78.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.5	21.5	17.6	DB	10.7	89.1	4.0	3.0	7.8	91.3	5.5	93.1	3.3	94.7	0.4	96.9	
			WB	10.2	80.9	3.8	2.2	7.4	82.5	5.2	83.8	3.0	85.0	0.2	86.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	35.8	39.4	44.0	51.2	60.4	69.6	72.6	73.1	66.7	56.3	47.3	38.8	(6)
	DBStd	14.58	7.75	7.36	6.30	5.77	4.99	4.35	3.16	4.00	5.69	6.21	8.60	7.76	(7)
	HDD50	1528	442	304	207	53	1	0	0	0	0	19	149	353	(8)
	HDD65	4526	906	718	650	415	156	11	1	2	49	275	531	812	(9)
	CDD50	3238	1	6	22	88	324	588	701	716	502	215	69	6	(10)
	CDD65	761	0	0	0	0	14	149	237	253	102	6	0	0	(11)
	CDH74	4224	0	0	3	2	30	640	1404	1811	326	8	0	0	(12)
	CDH80	454	0	0	0	0	3	70	113	244	24	0	0	0	(13)
	WSAvg	7.4	7.2	7.7	8.0	7.1	7.0	6.7	6.8	7.4	8.0	8.3	7.6	7.4	(14)
(15) Precipitation	PrecAvg	18.2	1.5	0.9	1.3	1.3	1.8	1.8	1.5	1.2	2.1	1.9	1.4	1.5	(15)
	PrecMax	31.6	4.5	2.3	3.6	4.4	4.2	6.8	5.2	3.8	6.2	6.5	3.3	4.9	(16)
	PrecMin	9.8	0.1	0.1	0.0	0.1	0.3	0.1	0.0	0.1	0.2	0.1	0.1	0.2	(17)
	PrecStd	4.9	1.1	0.6	0.8	1.0	1.1	1.5	1.2	1.1	1.6	1.4	0.8	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	62.5	69.2	72.2	77.5	84.8	84.5	87.2	83.5	74.8	69.2	59.0	(19)
		MCWB	49.1	53.8	52.0	58.0	65.2	70.9	72.7	76.6	65.0	67.0	60.3	53.5	(20)
	2%	DB	50.3	56.1	61.3	65.6	73.1	81.8	82.5	84.1	78.8	70.2	64.0	55.8	(21)
		MCWB	47.0	50.5	50.7	56.4	64.5	73.4	73.4	76.4	70.6	64.2	59.0	52.3	(22)
	5%	DB	47.4	51.3	56.1	62.1	70.4	79.9	81.1	82.4	77.2	67.3	61.3	52.5	(23)
		MCWB	45.4	46.9	48.6	55.6	64.2	72.6	73.5	74.8	71.1	62.7	58.6	49.8	(24)
	10%	DB	45.3	48.2	51.7	59.0	68.4	77.7	80.0	81.1	75.6	65.1	59.1	49.6	(25)
		MCWB	43.9	45.0	46.9	54.6	63.2	71.2	73.2	74.1	70.1	61.1	56.9	47.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	55.3	53.1	60.8	69.5	78.3	79.0	81.7	76.5	70.0	62.9	55.7	(27)
		MCDB	54.5	60.6	61.4	65.8	72.6	80.6	80.7	84.6	77.7	72.6	64.6	57.8	(28)
	2%	WB	47.7	51.2	51.5	58.7	67.6	76.3	77.3	78.7	74.9	66.1	61.1	53.1	(29)
		MCDB	49.1	56.1	59.2	63.4	70.8	79.4	79.9	81.7	76.5	68.0	62.6	54.7	(30)
	5%	WB	45.8	47.9	49.7	56.9	65.6	74.4	76.0	77.1	73.2	64.2	59.1	50.1	(31)
		MCDB	47.0	50.4	54.2	61.0	68.9	77.7	79.2	81.0	75.0	66.3	61.4	52.2	(32)
	10%	WB	44.2	45.7	48.0	54.8	63.9	72.6	74.6	75.6	71.8	62.3	56.6	47.4	(33)
		MCDB	45.2	47.3	50.7	58.1	67.3	76.0	78.4	79.9	74.4	64.4	58.7	49.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	10.3	11.3	10.9	11.2	12.2	12.7	12.8	12.7	11.4	10.2	10.9	(35)
		MCDBR	12.2	16.0	19.2	16.3	16.0	15.3	14.6	13.8	14.1	13.8	12.6	13.7	(36)
		MCWBR	9.1	10.7	10.4	8.9	9.9	9.5	10.2	11.3	10.5	9.5	8.7	10.2	(37)
	5% WB	MCDBR	10.3	15.0	15.0	13.5	13.2	12.6	12.4	12.2	11.5	11.9	10.6	12.8	(38)
		MCWBR	8.6	10.5	9.3	9.0	9.9	10.2	10.2	11.8	10.0	9.8	8.2	9.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.357	0.386	0.417	0.456	0.429	0.431	0.441	0.451	0.430	0.399	0.376	0.358	(40)
	taud		2.319	2.239	2.183	2.094	2.258	2.291	2.273	2.237	2.265	2.318	2.321	2.322	(41)
	Ebn at Noon		237	250	258	257	268	267	263	256	251	243	227	223	(42)
	Edh at Noon		27	34	41	49	42	41	41	42	37	31	27	24	(43)
(44) All-Sky Solar Radiation	RadAvg		413	657	1066	1541	1956	2137	2128	1905	1364	851	491	369	(44)
	RadStd		63	83	110	149	112	126	113	95	109	78	62	58	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	-221	+191	+117

Nomenclature: See separate page